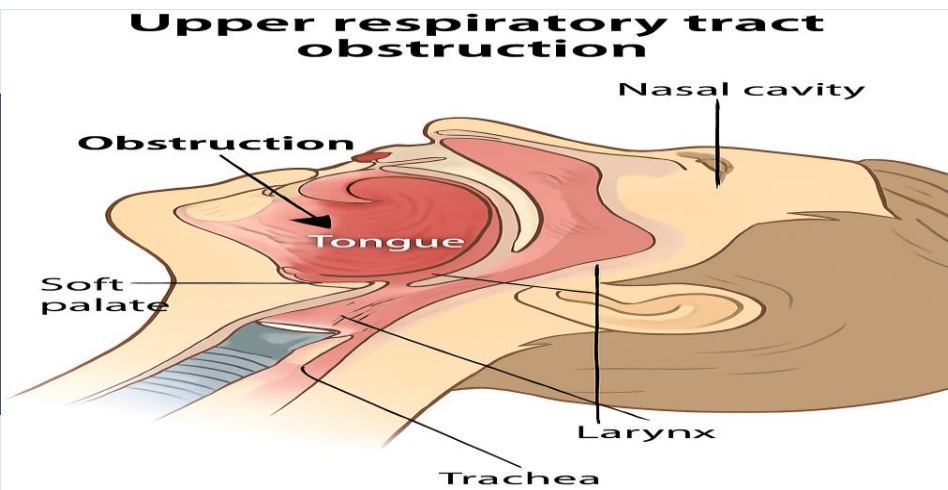


Upper respiratory tract obstruction

DR/ Zinab Mohamed



PrepLadder
RESIDENCY

**UPPER AIRWAY
OBSTRUCTION**

ENT Residency
LARYNX

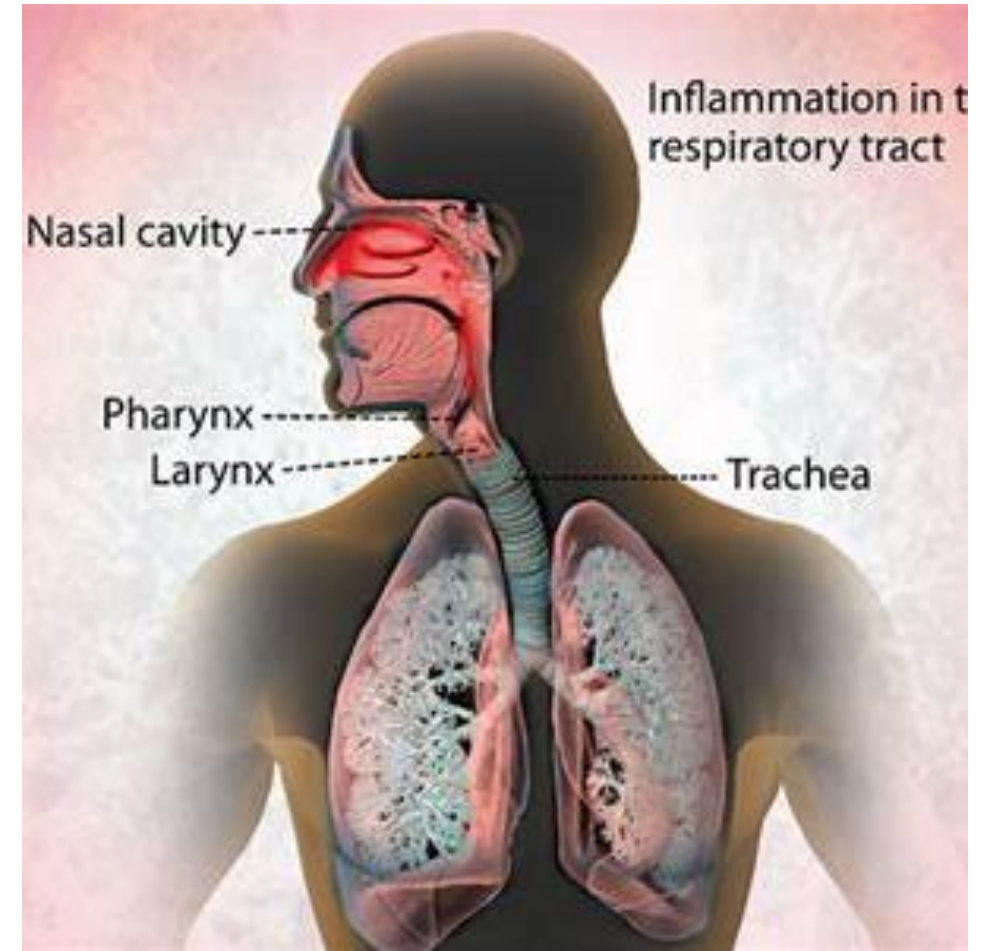


Agenda

- Introduction of Upper Respiratory Tract Obstruction
- Definition, causes of Upper Respiratory Tract Obstruction
- Symptoms of Upper Respiratory Tract Obstruction
- Assessment and diagnostic findings of Upper Respiratory Tract Obstruction
- Complications of Upper Respiratory Tract Obstruction
- Medical Management of Upper Respiratory Tract Obstruction
- Nursing care plan for patients with Upper Respiratory Tract Obstruction

- Upper respiratory tract obstruction refers to a partial or complete blockage of the airway above the vocal cords (larynx), including the **nose, pharynx, and larynx**, which can interfere with normal breathing.

- This can be **acute (sudden onset, potentially life-threatening) or chronic (develops over time).**



Pathophysiology of Upper Respiratory Tract Obstruction:

Anatomical narrowing or blockage due to swelling, mass, foreign body, trauma, or infection.

Any obstruction increases the resistance to airflow, particularly during inspiration

Obstruction can prevent adequate oxygen from reaching the alveoli, leading to hypoxia

High-pitched, noisy breathing (stridor) often results from turbulent airflow through a narrowed upper airway.

Prolonged obstruction leads to fatigue, respiratory failure, and potentially cardiac arrest.

Causes

1- Infections

- **Tonsillitis, pharyngitis**
(severe swelling)
- **Epiglottitis** (inflammation of the epiglottis)
- **Laryngitis**

2- Trauma

- Facial or neck injuries
- Inhalation injury (burns, smoke inhalation)
- Foreign body aspiration

3- Congenital abnormalities

- Choanal atresia
- Laryngomalacia

4- Neoplasms

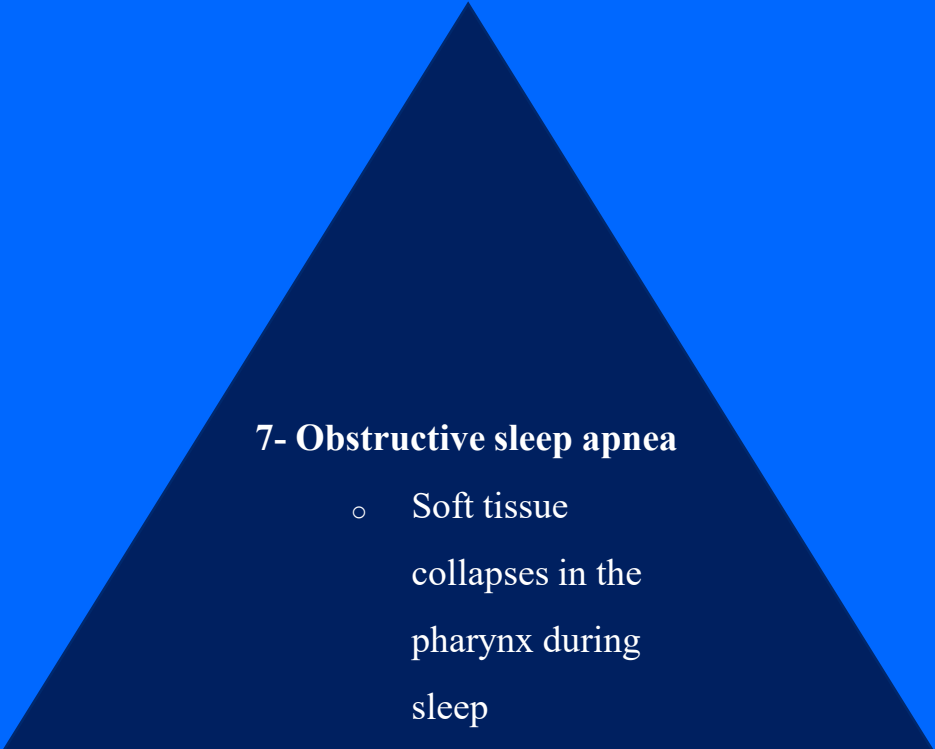
Tumors or growths in the airway (benign or malignant)

5- Allergic reactions

Angioedema or anaphylaxis causing swelling of airway structures

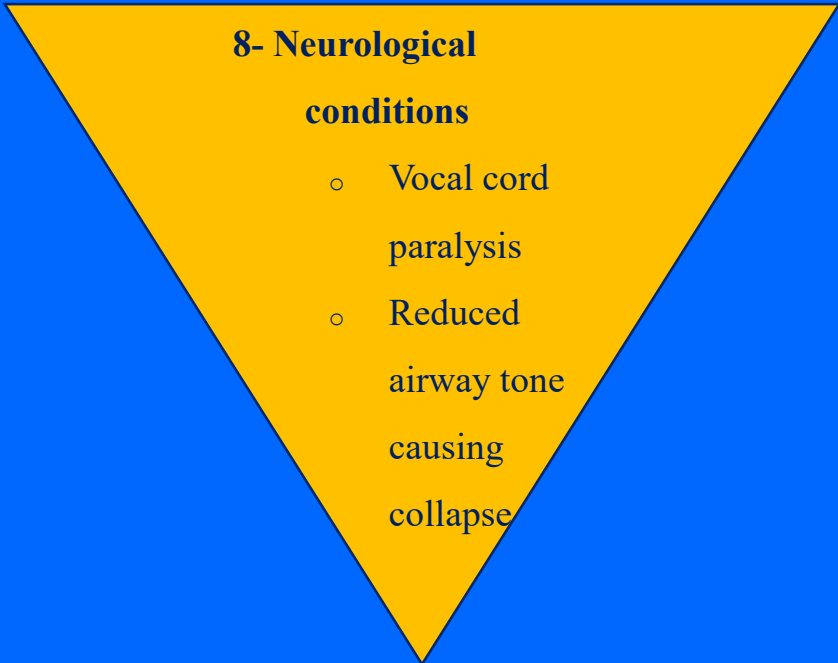
6- Foreign bodies

Objects lodged in the airway, especially common in children



7- Obstructive sleep apnea

- Soft tissue collapses in the pharynx during sleep



8- Neurological conditions

- Vocal cord paralysis
- Reduced airway tone causing collapse

Symptoms

Stridor

Dyspnea

Cough

Cyanosis

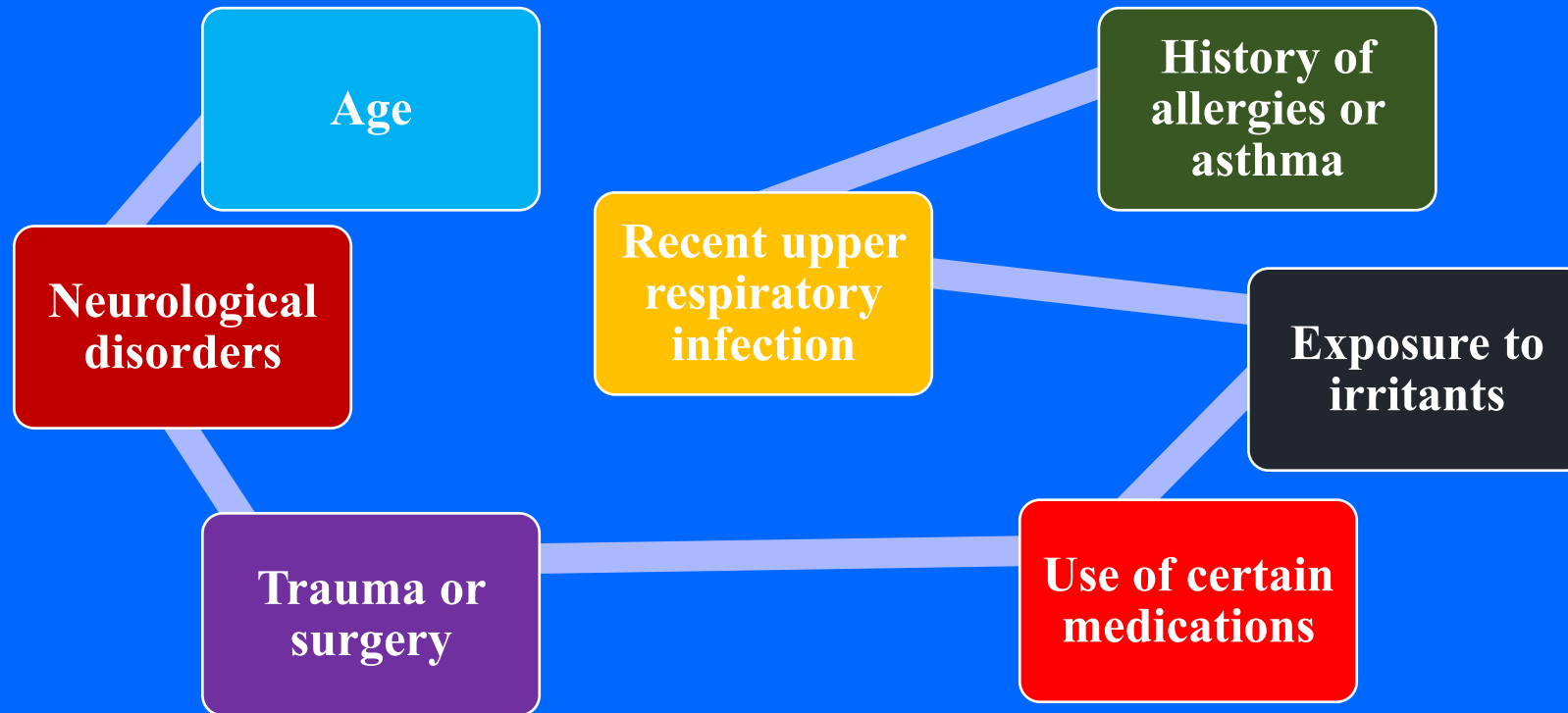
Hoarseness

Difficulty
swallowing

Restlessness

Tachypnea

Risk Factors



Types of Upper Respiratory Tract Obstruction:

1. Acute Obstruction

- Sudden onset; requires urgent management.

Causes:

- Foreign body aspiration (common in children)
- Acute allergic reaction (anaphylaxis → laryngeal edema)
- Infections (e.g., epiglottitis, croup, peritonsillar abscess)
- Trauma (e.g., facial fractures, burns, laryngeal injury)

2. Chronic Obstruction

Tumors (laryngeal,
nasopharyngeal)

Enlarged tonsils

Vocal cord paralysis

Congenital
abnormalities

3. Partial Obstruction

Airflow is reduced but still present.

Symptoms: hoarseness, dyspnea, cough.

4. Complete Obstruction

Total blockage of airflow; a medical emergency.

Symptoms: Inability to speak or breathe, cyanosis, loss of consciousness.

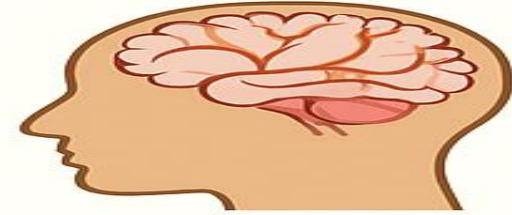
5. Extrinsic vs. Intrinsic Obstruction

Intrinsic: Caused by something within the airway (e.g., **tumor**).

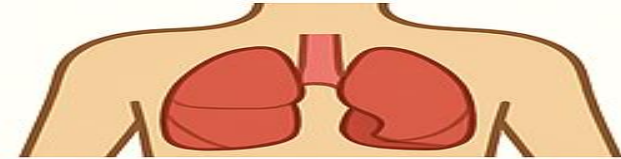
Extrinsic: Caused by external compression (e.g., thyroid mass pressing on trachea).

COMPLICATIONS

Hypoxia → brain damage



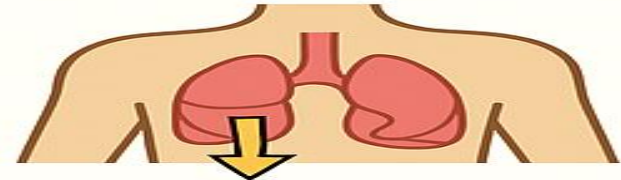
Respiratory arrest



Cardiac arrest



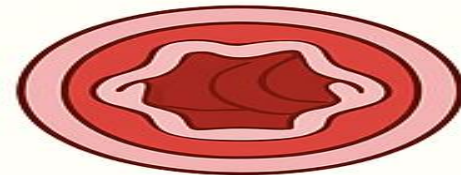
Aspiration pneumonia



Death (if untreated)



Airway edema



PREVENTION

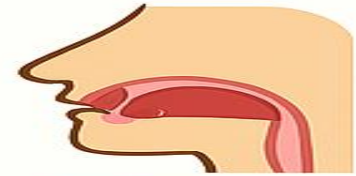
Vaccinations (e.g., Hib vaccine to prevent influenza)



Avoid choking hazards (especially in children)



Early treatment of infections (e.g. tonsillitis)



Proper management of allergies and asthma



Avoid smoke and pollutants



Use protective gear in high-risk jobs (e.g. chemical exposure)



TREATMENT

Treatment depends on cause and severity:

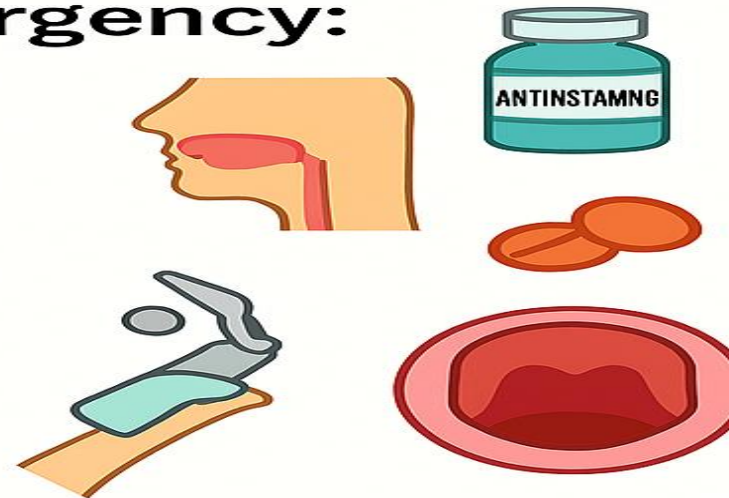
1. Medical:

- Oxygen therapy
- Corticosteroids
- Antibiotics
- Antihistamines



2. Surgical/Emergency:

- Endotracheal intubation
- Tracheostomy



ASSESSMENT

Treatment depends on cause and severity:

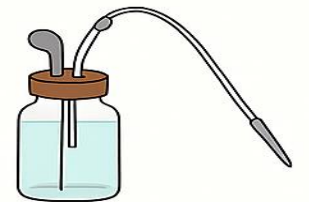
1. Medical:

- Monitor airway patency, respiratory rate, rhythm, and effort
- Check for cyanosis, use of accessory muscles
- Assess oxygen saturation (SpO_2)
- Observe for anxiety or altered mental status
- Take detailed history (onset, allergies, possible foreign body, infections)



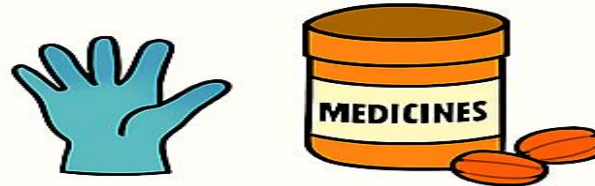
INTERVENTIONS

- Maintain a patent airway
- Position the patient upright
- Administer oxygen as ordered
- Prepare for emergency airway management
- Administer medications
- Monitor vital signs and SpO_2 frequently
- Provide a calm environment
- Encourage fluid intake if allowed
- Suction if needed



EDUCATION

- Teach signs of worsening obstruction
- Educate on allergy avoidance, infection control
- Prepare for emergency airway management
- Administer medications
- Provide a calm environment



Upper Respiratory Tract Infection



Respiratory Tract Infection

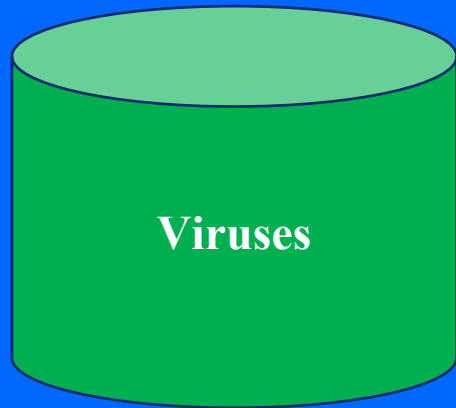
Respiratory Tract Infection refers to infections that affect the respiratory tract, which includes the nose, throat, airways, and lungs.

Upper Respiratory Tract

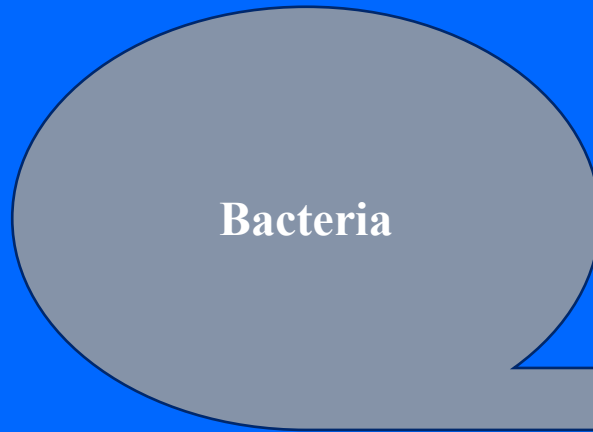
Infection (URTI): Involves nose, sinuses, pharynx, larynx (e.g., common cold, sinusitis, pharyngitis).

- **Lower Respiratory Tract Infection (LRTI):** Involves trachea, bronchi, lungs (e.g., bronchitis, pneumonia).

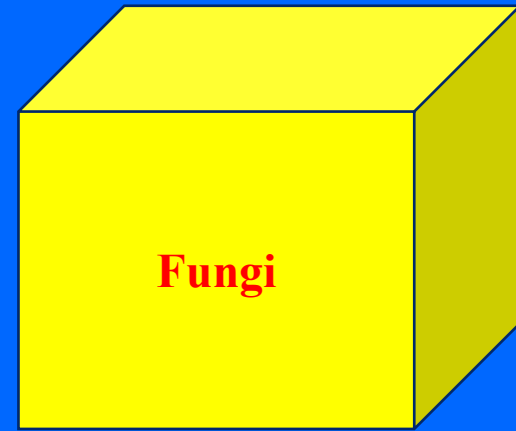
Causes



Viruses



Bacteria



Fungi

Others: Allergens and pollutants

Signs and Symptoms

General Symptoms



Upper RTI

- Fever, chills
- Malaise

- Sneezing
- Sore throat

- Nasal congestion, runny nose,
- Sneezing, sore throat, cough hoarse-

Severe Cases

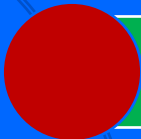


- Persistent cough (may produce sputum)
- Chest pain, shortness of breath, wheezing

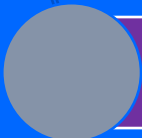
- Cyanosis
- Rapid breathing

- High fever
- Rapid breathing
- Confusion

Prevention



Regular handwashing with soap and water



Cover mouth and nose while coughing or sneezing.



Avoid close contact with infected individuals.



Use masks during outbreaks.

- . Avoid smoking and exposure to pollutants.
- . Vaccinations (e.g., influenza vaccine).
- . Maintain good nutrition and hydration.

Diagnosis

History and Physical Examination



Throat/nasal swab



Complete blood count (CBC)



Chest X-ray

+ Pulse oximetry



Treatment

Bacterial infections:
Appropriate antibiotics

Viral infections:
Mainly supportive care (rest, fluids, antipyretics)

Symptomatic treatment:

- Antipyretics
- Cough suppressants

Oxygen therapy

Hospitalization:
For severe infections

Nursing Care

- Monitor vital signs regularly
- Encourage adequate fluid intake
- Maintain airway patency
- Administer prescribed medications
- Promote rest and comfort.
- Educate patient and family about infection control measures.
- Assist with respiratory therapies (e.g. inhalation therapy).



Throat/nasal swab



Maintain airway patency



Educate patient and family about



+ Pulse oximetry



Thank
You